

4. A student investigated changes that happened when pairs of colourless solutions were mixed. The table shows the observations made after mixing.

Pair of solutions	Appearance of the reactants	Appearance when mixed	Temperature change when mixed
A	two colourless solutions	no change	increase
B	two colourless solutions	bubbles form	increase
C	two colourless solutions	white precipitate forms	no change
D	two colourless solutions	no change	no change

- (a) Which pair of solutions do **not** react? Give a reason for your answer.

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- (b) (i) The gas given off when pair **B** react is carbon dioxide. Describe the test for carbon dioxide gas.

[1]

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- (ii) One of the solutions in pair **B** is sodium carbonate. Give the formula of sodium carbonate.

[1]

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- (c) (i) Which pair of solutions is silver nitrate and sodium chloride?

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- (ii) What colour would sodium chloride give in a flame test?

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